

Stage 1 Modeling

Tree morphology measurement
 $\mathcal{G} = (\mathcal{B}, \mathcal{E})$

Branch material tests
→ priors $p(\theta)$

Fruit distribution model
(linear arc-length density)

Assemble FE operators
 $\mathbf{M}, \mathbf{K}(\theta), \mathbf{C}(\alpha, \beta)$

**Prior-constrained
FE model**

Stage 2 Calibration

Hammer test
(force + accel. signals)

H_1 FRF estimate
 $H_1 = G_{af}/G_{ff}$

Peak picking
 $\hat{f}_r, |\hat{H}(\omega_i)|$

MCMC calibration (emcee)
 $\theta \in \mathbb{R}^8$

Posterior
 $p(\theta | \hat{f}, \hat{H})$

Stage 3 Validation

Fixed-frequency
excitation test

Forward solve at $\hat{\theta}$
(posterior median)

Posterior predictive FRF
(NRMSE, peak error)

90% CI coverage check
vs. independent data

**Validated
calibrated model**

Stage 4 Recommendation

(p, f, A) sweep
 $\mathcal{P}_0 \times \mathcal{F} \times \mathcal{A}$

Multi-clamp Pareto
 C_{cov} vs σ_{tr}

Knee selection
 $\min \|(\tilde{y}_1, \tilde{y}_2)\|_2$

Posterior + Sobol
90% CI, $S_{T,i}$

Recommendation
 $(p^\dagger, f^\dagger, A^\dagger) \pm \text{CI}$